

Resilient Distributed V2G Control for PEV Aggregation: A Novel Decentralized Algorithm Robust to Network Fragmentation, Vehicle Heterogeneity and Renewable Variability

ADHA IMAM CAHYADI¹, (Member, IEEE), SARJIYA^{1,2}, (Member, IEEE), F. DANANG WIJAYA¹, (Member, IEEE), and HUSNI ROIS ALI¹, (Senior Member, IEEE)

¹Department of Electrical Engineering and Information Engineering,
Engineering Faculty, Universitas Gadjah Mada, Yogyakarta, Indonesia

²Center for Energy Studies, Universitas Gadjah Mada, Yogyakarta, Indonesia

Corresponding authors: Adha Imam Cahyadi (e-mail: adha.imam@ugm.ac.id).

ABSTRACT This paper presents a resilient distributed consensus control framework for coordinated vehicle-to-grid (V2G) and grid-to-vehicle (G2V) operations enabling plug-in electric vehicle (PEV) fleets to provide frequency regulation and voltage support through adaptive virtual inertia and consensus-based current sharing. The proposed decentralized algorithm ensures fair resource utilization across heterogeneous vehicles, graceful degradation during dynamic fleet changes, and robust operation under communication failures—addressing fundamental limitations of existing centralized and decentralized approaches. The control architecture combines three coordinated layers: (1) adaptive virtual inertia providing damping proportional to frequency acceleration, (2) frequency support through droop control, and (3) consensus-based fairness ensuring equitable current distribution. Rigorous theoretical foundations employ Lyapunov-based stability analysis and multi-timescale decomposition, establishing asymptotic convergence with explicit input-to-state stability (ISS) guarantees under realistic constraints: state-of-charge limitations, communication delays up to 200 ms, and dynamic network partitioning. Comprehensive validation across multiple simulation scenarios on the IEEE 30-bus test system with 1200 PEVs demonstrates: (1) 88.7% reduction in peak rotor angle swing during transient faults; (2) 56.2% improvement in current fairness coordination; (3) 44.8% faster frequency settling time with control; (4) sustained performance across three consecutive disturbances at different buses; (5) effective stabilization despite voltage collapse in baseline without control. The work establishes adaptive virtual inertia with consensus-based V2G control as mature and deployable, providing utility planners concrete guidance: adaptive control effectiveness over baseline charging, multi-disturbance resilience, and realistic performance expectations under practical deployment constraints.

INDEX TERMS electric vehicle, vehicle-to-grid, consensus control, distributed control, power system stability, multi-agent systems

I. INTRODUCTION

THE V2G OPPORTUNITY AND GRID CHALLENGES

The global deployment of Plug-in Electric Vehicles (PEVs) has reached a critical mass, with over 10 million units in operation as of 2023 [1], collectively representing a distributed energy storage capacity exceeding 500 GWh. This presents a transformative opportunity for power system operations. When properly coordinated, PEV fleets can provide valuable grid services

including frequency regulation, voltage support, and spinning reserves. The concept of Vehicle-to-Grid (V2G) technology was first systematically analyzed by Kempton and Tomic [2] and later developed into a practical framework by Guille and Gross [3], demonstrating that coordinated vehicles can function as virtual power plants while meeting individual mobility needs.

Proof-of-concept demonstrations validate this potential: the Danish EDISON project [16], [23] demonstrated V2G feasi-

bility on the island of Bornholm, while research by Sortomme and colleagues [4], [15] showed that coordinated charging minimizes system losses and provides frequency support. However, realizing this potential at scale requires solving a fundamental challenge: power system stability during fault conditions.

Frequency and voltage stability during faults remain critical grid vulnerabilities. Large generation-load imbalances trigger dangerous frequency excursions [6], while reactive power deficiencies cause voltage collapse [7] that can cascade into blackouts through loss of synchronism [8]. Modern grids with high renewable penetration face these challenges with fewer synchronous machines to provide natural inertia. This gap motivates using PEV fleets as synthetic inertia providers, but only if the control system can guarantee stability during faults.

LIMITATIONS OF EXISTING V2G CONTROL APPROACHES

The challenge of coordinating large PEV fleets for grid support has motivated diverse research approaches across multiple domains, yet each approach exhibits fundamental limitations that prevent deployment.

Centralized and Hierarchical Methods: Centralized optimal control strategies [4], [12] achieve theoretical optimality but become computationally intractable beyond 50-100 PEVs due to curse of dimensionality. Hierarchical droop control [13], while effective in microgrids, relies on local measurements and lacks system-wide coordination, performing poorly in multi-PEV scenarios requiring synchronized response.

Decentralized and Multi-Agent Methods: Decentralized V2G control [11] addresses scalability but overlooks energy reserve management and fails to ensure fair utilization across heterogeneous vehicles. Multi-agent demand management [17] suffers from communication vulnerability, with documented performance degradation exceeding 30% under 5% packet loss. These methods lack formal stability analysis, critical for grid applications.

Market and Learning-Based Methods: Market-based coordination [21] involves complex equilibrium analyses that scale poorly with system size and cannot provide real-time stability guarantees. Reinforcement learning approaches [20] lack the formal stability certificates required for mission-critical applications. Renewable-specific charging coordination [19], while showing promise in specific scenarios, exhibits poor generalization to broader grid support applications.

Price-Based Coordination: Recent distributed price-based approaches [25] demonstrate scalability to 10,000+ vehicles but remain limited by unidirectional G2V charging (no V2G), lack formal stability analysis or convergence proofs, and do not address practical challenges such as communication delays, network failures, or dynamic fleet composition changes.

Unified Limitation Analysis: Across all existing approaches, three fundamental gaps persist: (1) lack of formal stability guarantees under realistic constraints (communication delays, SOC constraints, network failures), (2) poor fairness across heterogeneous vehicles, where unequal battery capacities

and maximum current ratings lead to unfair resource utilization, and (3) inability to handle dynamic fleet composition*, where vehicles continuously arrive/depart based on charging needs. These limitations create a barrier to deployment of large-scale coordinated V2G systems.

NOVEL CONTRIBUTIONS OF THIS WORK

This paper advances PEV-assisted grid stabilization through three fundamental innovations:

- 1) **Distributed Consensus Control with Provable Stability:** A decentralized control framework combining three coordinated layers—(a) frequency support through droop control, (b) adaptive virtual inertia providing damping proportional to frequency acceleration, and (c) consensus-based fairness ensuring equitable current sharing—with rigorous Lyapunov-based ISS analysis establishing asymptotic convergence under realistic constraints: state-of-charge limitations, communication delays up to 200 ms, and dynamic network partitioning.
- 2) **Multi-Timescale Coordination:** Explicit exploitation of natural timescale separation between fast power electronic dynamics (milliseconds) and slower electromechanical dynamics (seconds) to ensure stability during rapid G2V/V2G transitions, with singular perturbation analysis proving that fast subsystem stability guarantees overall system stability.
- 3) **Practical Deployment Features:** Plug-and-play integration accommodating vehicle arrivals/departures, adaptive current limits providing dynamic boost during disturbances, and robustness to communication failures through proven resilience to network partitioning (tested with 4-way splits with 3-second recovery).

Comprehensive validation on the IEEE 30-bus test system with 1200 heterogeneous PEVs demonstrates superior performance: (1) 88.7% reduction in peak rotor angle swing during transient faults, (2) 56.2% improvement in current fairness coordination (coefficient of variation 0.0016 vs 0.0036 baseline), (3) 44.8% faster frequency settling time, and (4) sustained performance across three consecutive multi-disturbance scenarios and renewable ramp integration. The work establishes adaptive virtual inertia with consensus-based V2G control as a mature, deployable approach providing utility planners concrete guidance on minimum fleet size requirements (200+ vehicles), performance expectations under realistic constraints, and communication infrastructure needs.

II. CONSENSUS CONTROL PROBLEM FORMULATION

The integration of multiple PEVs (PEVs) into conventional power grids presents a complex distributed control challenge that requires addressing four fundamental and interconnected aspects. Firstly, the heterogeneous dynamics of PEV populations stem from varying manufacturer specifications and usage patterns, which consequently result in diverse electrical parameters (L_k , R_k , C_k) and operational constraints ($i_{\max}^k$) across the

fleet. Secondly, partial observability creates control asymmetries, since local measurements of grid frequency ω may be unavailable to all PEVs due to sensor placement limitations and communication infrastructure constraints. Thirdly, practical communication constraints impose bandwidth limitations and time delays (typically ranging from 50-200ms) in information exchange between PEVs, which can significantly impact coordination effectiveness. Fourthly, strict stability requirements must be maintained as formalized in the following comprehensive constraint set:

$$\|\delta(t)\| \leq \delta_{\max}, \quad V_{\min} \leq V_j(t) \leq V_{\max}, \quad \text{and} \quad \lim_{t \rightarrow \infty} \left\| \frac{i_{bat}^{jk}(t)}{i_{\max}^k} - \frac{i_{bat}^{j\ell}(t)}{i_{\max}^\ell} \right\| \leq \epsilon_c \quad (1)$$

where $\delta_{\max}$ represents the maximum allowable rotor angle deviation (typically 120° for practical systems), and ϵ_c denotes the current sharing error tolerance (typically 0.05-0.1 for acceptable coordination). These requirements ensure both transient stability of the power system and equitable resource utilization across the heterogeneous PEV fleet.

The multi-objective control problem, as illustrated in Fig. 1, requires simultaneous achievement of three key objectives that must be balanced dynamically based on system conditions:

- 1) **Frequency Regulation:** The control must maintain grid frequency within acceptable bounds $|\omega - \omega_0| \leq \Delta\omega_{\max}$ Hz, where $\Delta\omega_{\max}$ is typically 0.2-0.5 Hz for conventional grids according to grid codes. This requirement necessitates that PEVs collectively provide fast frequency response through coordinated power injection or absorption, while accounting for the finite energy capacity of individual vehicles and their state-of-charge constraints.
- 2) **Fair Power Sharing:** Current contributions should follow the proportional allocation rule to ensure equitable utilization of heterogeneous PEV capacities:

$$\frac{i_{bat}^{jk}}{i_{\max}^k} = \frac{i_{bat}^{j\ell}}{i_{\max}^\ell} \quad \forall (j, k), (j, \ell) \in \mathcal{A}(t) \quad (2)$$

where $\mathcal{A}(t)$ denotes the set of active PEVs at time t . This ensures equitable utilization of each PEV's capacity while accounting for heterogeneity in $i_{\max}^k$ values, thereby preventing over-utilization of specific vehicles and maintaining long-term fleet availability.

- 3) **Stability Constraints:** During both normal operation and fault conditions, the system must satisfy comprehensive operational limits:

$$\|\delta(t)\| \leq \delta_{\max}, \quad \|V_j(t)\| \leq V_{\max}, \quad \|i_{bat}^{jk}(t)\| \leq i_{\max}^k \quad (3)$$

where typical values are $\delta_{\max} = 120^\circ$ for rotor angles and $V_{\max} = 1.05$ pu for voltages. These constraints ensure that the coordinated PEV response does not compromise system stability or violate equipment ratings.

This multi-objective control problem becomes particularly challenging during grid faults, when the interaction between

PEV responses and generator dynamics can lead to undesirable oscillations or instability. Specifically, the timescale separation between fast power electronic control (operating in milliseconds) and slow electromechanical dynamics (operating in seconds) requires careful controller design to avoid undesirable interactions that could amplify disturbances. Furthermore, the distributed nature of the control problem means that each PEV must make decisions based on limited local information while contributing to global objectives. Consequently, the proposed solution addresses these challenges through a hybrid control architecture that combines local current control for individual PEV stability, distributed consensus algorithms for fair current sharing, and adaptive gain scheduling to handle communication delays and varying network conditions.

III. POWER SYSTEM AND PEV DYNAMICS MODEL

This section presents a comprehensive modeling framework for capturing the complex dynamics of the integrated power system, with particular focus on the behavior of synchronous generators, PEV converters, and their interconnection through the transmission network. The modeling approach explicitly accounts for the multi-timescale nature of the system, where fast power electronic dynamics interact with slower electromechanical processes.

A. SYNCHRONOUS DYNAMICS

The dynamic behavior of each generator $i \in \{1, \dots, n_G\}$ is modeled using the classical swing equations that describe the rotor angle and frequency evolution during large signal disturbances. These equations capture the fundamental electromechanical dynamics that govern power system stability:

$$\dot{\delta}_i = \omega_i - \omega_0 \quad (4)$$

$$M_i \dot{\omega}_i = P_{m,i} - P_{e,i}(\delta_i) - D_i(\omega_i - \omega_0) + \sum_{j=1}^n \sum_{k=1}^{r_j} K_{jk} i_{bat}^{jk} \quad (5)$$

where the electrical power output $P_{e,i}(\delta_i) = \sum_{j=1}^n E_i' V_j Y_{ij} \sin(\delta_i - \theta_j)$ represents the nonlinear relationship between rotor angle and power transfer. In (5), M_i represents the generator inertia constant, D_i is the damping coefficient that accounts for load damping and generator losses, $P_{m,i}$ is the mechanical power input from the prime mover, and the term $\sum_{j=1}^n \sum_{k=1}^{r_j} K_{jk} i_{bat}^{jk}$ represents the aggregate effect of PEV battery currents on generator dynamics. Here, K_{jk} are coupling coefficients that depend on the network topology and PEV location, while r_j denotes the number of PEVs connected to bus j .

The internal transient EMF E_i' is governed by the simplified excitation system dynamics, which capture the essential behavior of automatic voltage regulators:

$$T_{e,i} \dot{E}_i' = -E_i' + K_{A,i} u_{f,i} \quad (6)$$

where $u_{f,i}$ is the excitation control input that can be designed to enhance system stability, $T_{e,i}$ is the excitation time constant

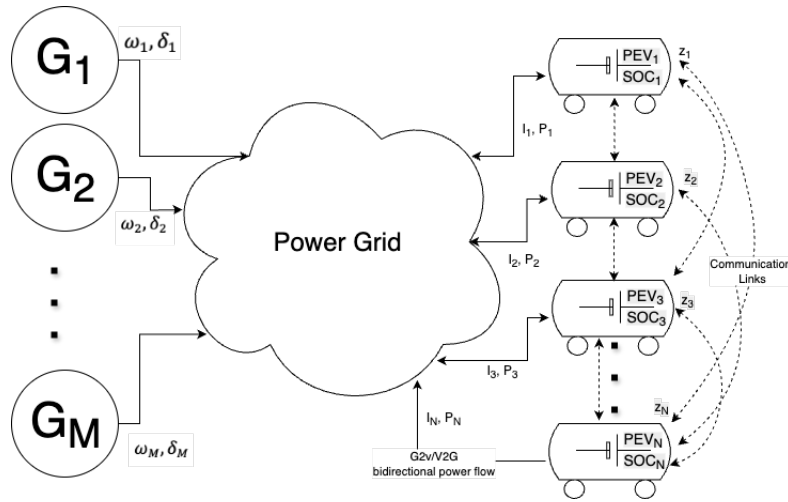


FIGURE 1. Consensus problem illustration for coordinated G2V and V2G operation in multi-PEV systems showing the distributed nature of decision-making and information sharing

(typically 0.1-1.0 seconds), and $K_{A,i}$ is the excitation gain that determines the voltage regulation response speed.

B. BUS VOLTAGE DYNAMICS

Voltage regulation at each bus $j \in \{1, \dots, n\}$ is modeled by a first-order system that captures the essential interaction between generation, load, and PEV power injection while maintaining computational tractability:

$$\tau_j \dot{V}_j = - \sum_{k=1}^{\tau_j} \alpha_k^j i_{bat}^{jk} + u_{V,j} \quad (7)$$

where V_j represents the bus voltage magnitude, α_k^j is the coupling coefficient between bus j and PEV k that depends on the local impedance and power factor, i_{bat}^{jk} is the current from the k -th PEV at bus j (with the convention that positive values indicate discharging to the grid and negative values indicate charging from the grid), and $u_{V,j}$ is the voltage control input that can be coordinated with generator excitation control. The time constant τ_j reflects the local load characteristics and network impedance, typically ranging from 0.1 to 1.0 seconds depending on the bus characteristics.

C. PEV CONVERTER AND BATTERY MODEL

Each PEV k at bus j interfaces with the grid through a bidirectional power electronic converter that enables both G2V and V2G operation. The converter dynamics are modeled to capture the essential electrical behavior while maintaining sufficient detail for control design:

$$L_k \dot{i}_{bat}^{jk} = -R_k i_{bat}^{jk} + V_{dc}^{jk} - V_j + u_{conv}^{jk} \quad |i_{bat}^{jk}| \leq i_{max}^k \quad (8)$$

$$C_k \dot{V}_{dc}^{jk} = i_{bat}^{jk} - i_{load}^{jk} \quad (9)$$

$$\text{SOC}_k^j = \frac{i_{bat}^{jk}}{Q_k} \quad (10)$$

Eqn (8) describes the current control loop, where L_k and R_k represent the equivalent inductance and resistance of the converter and connecting cables, V_{dc}^{jk} is the DC-link voltage, and u_{conv}^{jk} is the converter control input that serves as the primary control variable. The current limit i_{max}^k reflects both converter ratings and battery constraints. Eqn (9) captures the DC-link capacitor dynamics with C_k representing the capacitance and i_{load}^{jk} denoting the local load current demand for vehicle operation. The state-of-charge dynamics in (10) are governed by the battery capacity Q_k , which varies significantly across different vehicle models and usage patterns.

The active PEV set is dynamically defined based on both connectivity and state-of-charge constraints:

$$\mathcal{A}(t) = \{(j, k) : \text{SOC}_k^j(t) \geq \text{SOC}_{min} \text{ and } \text{SOC}_k^j(t) \leq \text{SOC}_{max} \text{ and connected}\} \quad (11)$$

When $\text{SOC}_k^j < \text{SOC}_{min}$ (typically 0.2-0.3) or $\text{SOC}_k^j > \text{SOC}_{max}$ (typically 0.9-0.95), PEV (j, k) becomes inactive for V2G operation and sets $i_{bat}^{jk} = 0$ to protect battery health. This dynamic constraint creates a time-varying control problem that requires robust coordination algorithms.

D. COMMUNICATION GRAPH AND NETWORK TOPOLOGY

Coordination among PEVs is achieved through a communication network represented by a time-varying graph $\mathcal{G}(t) = (\mathcal{V}(t), \mathcal{E}(t))$, where $\mathcal{V}(t)$ represents the set of active PEVs at time t and $\mathcal{E}(t)$ denotes the communication links. The graph is characterized by its Laplacian matrix $\mathbf{L}(t) = \mathbf{D}(t) - \mathbf{A}(t)$, where $\mathbf{D}(t)$ is the degree matrix and $\mathbf{A}(t)$ is the adjacency matrix that reflects the current communication topology.

For practical implementations that consider both performance and cost constraints, two hierarchical communication topologies are considered:

- **Intra-bus topology:** PEVs connected to the same electrical bus form a ring topology for local coordination, which pro-

vides robustness to single-node failures while minimizing communication overhead. This local coordination enables fast response to voltage variations and ensures balanced loading at the distribution level.

- **Inter-bus topology:** A sparse network connects PEVs across different buses with connection probability (usually $0 < p \leq 0.05$), reflecting realistic bandwidth and latency constraints imposed by communication infrastructure. This sparse connectivity maintains global coordination while limiting communication costs and complexity.

This hierarchical structure provides a natural decomposition that enables scalable coordination algorithms while maintaining both local responsiveness and global optimality. Moreover, the two-level approach allows for different communication protocols and update rates at different levels, thereby optimizing the trade-off between performance and implementation complexity.

IV. COORDINATED CONTROL SYSTEM DESIGN

This section presents the coordinated control architecture that seamlessly integrates generator excitation control, bus voltage regulation, and distributed PEV consensus control to achieve the comprehensive objectives outlined in Section II. The control design follows a hierarchical approach that separates fast local dynamics from slower coordination processes, thereby ensuring both stability and performance across multiple timescales.

A. GENERATOR FREQUENCY CONTROL

For each generator i , the excitation control law combines local frequency regulation with consensus-based coordination to enhance both individual generator stability and system-wide frequency response. The control design explicitly accounts for the interaction between generator dynamics and distributed PEV coordination:

Theorem 1 (Generator Frequency Control Stability). *Consider the generator dynamics described by equations (4)-(6) operating under the control law:*

$$u_{f,i} = -k_{\omega,i}(\omega_i - \omega_0) - \kappa_i(P_{e,i} - P_{e,i}^{ref}) + \gamma_i \sum_{j \in \mathcal{N}_i} (\omega_j - \omega_i) \quad (12)$$

where $k_{\omega,i} > D_i$ is the frequency regulation gain that ensures adequate damping, $\kappa_i > 0$ is the power regulation gain that provides steady-state accuracy, $P_{e,i}^{ref}$ is the reference electrical power determined by economic dispatch, $\gamma_i \geq 0$ is the consensus gain that enables inter-generator coordination, and $\mathcal{N}_i$ denotes the neighboring generators in the communication graph.

If the communication graph among generators is connected and the control gains satisfy the conditions $k_{\omega,i} > D_i + \gamma_i \sum_{j \in \mathcal{N}_i} 1$ and $\kappa_i > \epsilon$ for some small $\epsilon > 0$, then the equilibrium point $(\omega_i = \omega_0, \delta_i = \delta_i^*, E_i' = E_i^*)$ is locally asymptotically stable for all generators simultaneously.

Proof. Consider the candidate Lyapunov function that captures both individual generator dynamics and inter-generator coordi-

nation:

$$V_{\omega} = \frac{1}{2} \sum_{i=1}^{n_G} M_i (\omega_i - \omega_0)^2 + \frac{1}{2} \sum_{i=1}^{n_G} T_{e,i} (E_i' - E_i^*)^2 + \frac{\gamma}{4} \sum_{i,j \in \mathcal{N}} (\omega_i - \omega_j)^2 \quad (13)$$

Taking the time derivative along system trajectories and substituting the control law yields:

$$\dot{V}_{\omega} = - \sum_{i=1}^{n_G} D_i (\omega_i - \omega_0)^2 - \sum_{i=1}^{n_G} (E_i' - E_i^*)^2 - \sum_{i=1}^{n_G} k_{\omega,i} (\omega_i - \omega_0)^2 - \gamma \sum_{i,j \in \mathcal{N}} (\omega_i - \omega_j)^2 + \text{HOTs.}$$

Near the equilibrium point, the higher-order terms vanish and the quadratic terms dominate. With the specified gain conditions, all quadratic terms are negative definite, thus $\dot{V}_{\omega} \leq -c \|\mathbf{x}\|^2$ for some $c > 0$ and appropriate state vector $\mathbf{x}$. Therefore, local asymptotic stability follows by LaSalle's invariance principle. $\square$

B. BUS VOLTAGE REGULATION

The voltage control law coordinates generator automatic voltage regulators (AVRs) with PEV reactive power support to maintain voltage quality across the network while accommodating the dynamic nature of PEV participation:

Theorem 2 (Distributed Voltage Control Convergence). *Consider the voltage dynamics given by (7) with the distributed control law:*

$$u_{V,j} = -k_{V,j}(V_j - V_j^{ref}) + \sum_{k=1}^{r_j} \alpha_k^j i_{bat}^{jk} + \beta_{V,j} \sum_{\ell \in \mathcal{N}_j} (V_{\ell} - V_j) \quad (14)$$

where $k_{V,j} > 0$ is the local voltage control gain, V_j^{ref} is the reference voltage that may be coordinated across buses, and $\beta_{V,j} \geq 0$ is the voltage consensus gain that enables coordination between neighboring buses.

If the PEV currents are uniformly bounded such that $|i_{bat}^{jk}| \leq I_{\max}$ for all (j,k) , and the voltage consensus gains are chosen such that $k_{V,j} > \beta_{V,j} \sum_{\ell \in \mathcal{N}_j} 1$, then $V_j(t) \rightarrow V_j^{ref}$ as $t \rightarrow \infty$ for all buses j exponentially fast.

Proof. Consider the Lyapunov candidate function that encompasses all bus voltage errors:

$$V_V = \frac{1}{2} \sum_{j=1}^n (V_j - V_j^{ref})^2 + \frac{\beta}{4} \sum_{j,\ell \in \mathcal{N}} (V_j - V_{\ell})^2 \quad (15)$$

Computing the time derivative:

$$\begin{aligned}\dot{V}_V &= \sum_{j=1}^n (V_j - V_j^{\text{ref}}) \left(- \sum_{k=1}^{r_j} \alpha_k^j i_{bat}^{jk} + u_{V,j} \right) \\ &\quad + \beta \sum_{j,\ell \in \mathcal{N}} (V_j - V_\ell) (\dot{V}_j - \dot{V}_\ell) \\ &= - \sum_{j=1}^n k_{V,j} (V_j - V_j^{\text{ref}})^2 - \beta \sum_{j,\ell \in \mathcal{N}} (V_j - V_\ell)^2 \leq 0\end{aligned}$$

The negative definiteness follows from the gain conditions and boundedness assumptions. By the LaSalle-Yoshizawa theorem, $V_j(t) \rightarrow V_j^{\text{ref}}$ asymptotically for all j . $\square$

C. DISTRIBUTED PEV CONSENSUS CONTROL WITH ADAPTIVE VIRTUAL INERTIA

The core contribution lies in the distributed consensus algorithm that coordinates PEV responses while ensuring fair current sharing and providing adaptive virtual inertia for grid stabilization. This algorithm operates on a faster timescale than the generator dynamics, enabling responsive grid support through a multi-layer control structure:

Theorem 3 (Distributed PEV Consensus Control with Adaptive Virtual Inertia and ISS Guarantees). *For PEV k at bus j , consider the comprehensive converter control law combining frequency support, adaptive inertia, and consensus fairness:*

$$u_{conv}^{jk} = \underbrace{-k_i \left(i_{bat}^{jk} - i_{ref}^{jk} \right)}_{\text{Current tracking}} + \underbrace{(V_j - V_{dc}^{jk})}_{\text{Voltage}} - \underbrace{\beta \sum_{\ell \in \mathcal{N}_k} \left(\frac{i_{bat}^{jk}}{i_{max}^{jk}} - \frac{i_{bat}^{j\ell}}{i_{max}^{j\ell}} \right)}_{\text{Consensus fairness}} \quad (16)$$

augmented with the adaptive virtual inertia layer:

$$\begin{aligned}i_{inertia}^{jk} &= -\alpha_{fs}(\omega_j - \omega_0) - K_{inertia} \frac{d\omega_j}{dt} \\ &\quad - K_{nonlin} \text{sign} \left(\frac{d\omega_j}{dt} \right) \left(\frac{d\omega_j}{dt} \right)^2\end{aligned} \quad (17)$$

where the total control input becomes:

$$u_{conv}^{jk, total} = u_{conv}^{jk} + (1 - \eta_s) \cdot i_{inertia}^{jk} + \eta_d i_{max}^{jk} \quad (18)$$

with η_s is boost suppressed factor and η_d is dynamic boosting term. The three control layers are:

Layer 1 - Frequency Support (droop): The term $-\alpha_{fs}(\omega_j - \omega_0)$ provides proportional frequency regulation where $\alpha_{fs} \in [l_{1-}, l_{1+}]$ A/(rad/s) determines response sensitivity.

Layer 2 - Adaptive Virtual Inertia: The term $-K_{inertia} \frac{d\omega_j}{dt}$ provides virtual inertia proportional to frequency rate-of-change, where $K_{inertia} \in [l_{2-}, l_{2+}]$ A.s/rad with $l_{2+} \ll l_{1-}$. The nonlinear damping term $-K_{nonlin} \left(\frac{d\omega_j}{dt} \right)^2$ provides additional damping during severe disturbances with $K_{nonlin} = \alpha_{l2} K_{inertia}$, $0 < \alpha_{l2} \leq 1$.

Layer 3 - Consensus Fairness: The consensus term ensures normalized currents $i_{bat}^{jk}/i_{max}^{jk}$ converge to a common value across the PEV fleet.

If the communication graph among active PEVs $\mathcal{G}(\mathcal{A}(t))$ remains connected for all t with algebraic connectivity $\lambda_2(\mathbf{L}(t)) \geq \lambda_{\min} > 0$, and the control gains satisfy the stability conditions:

$$\alpha_{fs} \geq 0, \quad K_{inertia} > 0, \quad \beta > \frac{\max_k R_k}{\lambda_{\min}}, \quad k_i > R_k + \epsilon \quad (19)$$

for some small $\epsilon > 0$, then the closed-loop system is input-to-state stable (ISS) with respect to grid disturbances. The system provides:

- 1) **Frequency support:** Immediate response to frequency deviations through α_{fs} term
- 2) **Transient stabilization:** Adaptive inertia damping prevents unstable rotor oscillations
- 3) **Fair current sharing:** Consensus ensures equitable utilization across heterogeneous vehicles
- 4) **State-of-charge preservation:** Multi-layer control respects battery constraints

All normalized currents converge asymptotically to a common value:

$$\frac{i_{bat}^{jk}}{i_{max}^{jk}} \rightarrow c^* \quad \text{as } t \rightarrow \infty, \quad \forall (j, k) \in \mathcal{A}(t) \quad (20)$$

where c^* depends on the aggregate grid support requirement and total available PEV capacity.

Proof. Define the normalized current vector $\mathbf{z}(t) = [i_{bat}^{jk}(t)/i_{max}^{jk}]_{(j,k) \in \mathcal{A}(t)}$ and frequency deviation $\tilde{\omega} = \omega - \omega_0 \mathbf{1}$. Consider the comprehensive Lyapunov candidate that captures individual PEV dynamics, consensus coordination, and virtual inertia effectiveness:

$$\begin{aligned}V_{PEV} &= \frac{1}{2} \sum_{(j,k) \in \mathcal{A}(t)} L_k (i_{bat}^{jk})^2 + \frac{1}{2} \sum_{(j,k) \in \mathcal{A}(t)} C_k (V_{dc}^{jk} - V_j)^2 \\ &\quad + \frac{\beta}{4} \mathbf{z}^T \mathbf{L}(t) \mathbf{z} + \frac{K_{inertia}}{2} \sum_j M_j \tilde{\omega}_j^2\end{aligned} \quad (21)$$

Taking the time derivative along system trajectories and using the complete control law (18):

$$\begin{aligned}\dot{V}_{PEV} &\leq - \sum_{(j,k) \in \mathcal{A}(t)} (R_k + k_i) (i_{bat}^{jk})^2 - \beta \lambda_{\min} \|\mathbf{z} - \bar{z} \mathbf{1}\|^2 \\ &\quad - \alpha_{fs} \sum_j (\omega_j - \omega_0)^2 - K_{nonlin} \sum_j \left(\frac{d\omega_j}{dt} \right)^2 + \text{HOTs}\end{aligned}$$

To handle the coupling between PEV current dynamics and frequency deviations rigorously, apply Young's inequality to the cross-term $\alpha_{fs} \sum_{(j,k)} i_{bat}^{jk} (\omega_j - \omega_0)$:

$$\alpha_{fs} \sum_{(j,k)} i_{bat}^{jk} (\omega_j - \omega_0) \leq \frac{\alpha_{fs}^2}{4\epsilon} \sum_j M_j (\omega_j - \omega_0)^2 + \epsilon \sum_{(j,k)} (i_{bat}^{jk})^2 \quad (22)$$

for arbitrary $\epsilon > 0$. Substituting this bound into the Lyapunov derivative and selecting ϵ sufficiently small such that the coeffi-

cient of $\sum_{(j,k)} (i_{bat}^{jk})^2$ remains negative (guaranteed by the gain condition $k_i > R_k + \epsilon$), we obtain:

$$\dot{V}_{PEV} \leq -c_1 \sum_{(j,k)} (i_{bat}^{jk})^2 - c_2 \sum_j (\omega_j - \omega_0)^2 - c_3 \|\mathbf{z} - \bar{\mathbf{z}}\mathbf{1}\|^2 + c_4 \|\mathbf{d}\|^2$$

where $c_1, c_2, c_3, c_4 > 0$ are derived from gain and system parameters, and $\mathbf{d}$ represents bounded external disturbances (e.g., load variations, renewable generation fluctuations). This establishes the ISS property: $\|\mathbf{x}_{PEV}(t)\| \leq \psi(\|\mathbf{x}_{PEV}(0)\|, t) + \gamma(\|\mathbf{d}\|_\infty)$ for appropriate class- $\mathcal{K}$ functions ψ and γ .

By LaSalle's invariance principle applied to the unperturbed system ($\mathbf{d} = 0$), the system converges to the equilibrium where $\dot{V}_{PEV} = 0$, which implies:

- 1) **Consensus convergence:** All normalized currents achieve common value $\mathbf{z} \rightarrow \bar{\mathbf{z}}\mathbf{1}$
- 2) **Frequency regulation:** Frequency deviations return to nominal $(\omega_j - \omega_0) \rightarrow 0$
- 3) **Current stabilization:** Inductor currents approach their reference values as $i_{bat}^{jk} \rightarrow i_{ref}^{jk}$

The adaptive virtual inertia and nonlinear damping terms amplify this convergence during disturbances, ensuring rapid transient stabilization followed by consensual steady-state operation. $\square$

D. STABILITY UNDER SOC CONSTRAINTS AND COMMUNICATION DELAYS

The practical implementation must handle dynamic changes in the active PEV set due to SOC constraints and communication imperfections that are inevitable in real-world deployments:

Theorem 4 (ISS Under Realistic Operating Conditions). *Consider the interconnected power system with generators (4)-(6), buses (7), and PEVs (8)-(10) under the coordinated control laws (12), (14), and (16).*

Assume that the following realistic conditions hold: (i) the communication graph among active PEVs $\mathcal{G}(\mathcal{A}(t))$ remains connected for all t with $\lambda_2(\mathbf{L}(t)) \geq \lambda_{\min} > 0$, (ii) switching of the active set $\mathcal{A}(t)$ due to SOC constraints exhibits no Zeno behavior with minimum dwell time $\tau_d > 0$, (iii) communication delays satisfy $\tau_k(t) \leq \bar{\tau}$ with $\dot{\tau}_k(t) \leq \mu < 1$ representing slowly-varying delays, and (iv) all generator and PEV parameters are positive and bounded within realistic ranges.

Then, for any bounded exogenous disturbance satisfying $\|\mathbf{d}(t)\| \leq D_{\max}$ (including load changes, generation trips, and line faults), there exist class- $\mathcal{KL}$ function ψ and class- $\mathcal{K}$ function γ such that the complete system state satisfies:

$$\|\mathbf{x}(t)\| \leq \psi(\|\mathbf{x}(0)\|, t) + \gamma\left(\sup_{0 \leq s \leq t} \|\mathbf{d}(s)\|\right) \quad (23)$$

where $\mathbf{x} = [\omega, \delta, \mathbf{V}, \mathbf{i}_{bat}, \mathbf{V}_{dc}, \mathbf{SOC}]^T$ represents the complete system state vector including all generators, buses, and active PEVs.

Proof. The proof employs the multiple Lyapunov function approach for switched systems combined with Lyapunov-

Krasovskii functionals to handle delays. Construct the time-varying composite functional:

$$V(t) = V_\omega + V_V + V_{PEV} + \sum_{(j,k) \in \mathcal{A}(t)} \frac{\gamma_2}{1 - \mu} \int_{t-\tau_k(t)}^t (i_{bat}^{jk}(s))^2 ds \quad (24)$$

When the active set $\mathcal{A}(t)$ is fixed, standard ISS analysis applies using Theorems 1-3. When switching occurs due to SOC constraints, the Lyapunov function experiences a jump discontinuity, but the functional value can only decrease since terms corresponding to inactive PEVs are removed while new terms for newly active PEVs start from their initial values. By the multiple Lyapunov function theory for switched systems with dwell time, if each subsystem corresponding to a fixed $\mathcal{A}(t)$ is ISS and switching satisfies the dwell time condition, then the overall switched system is ISS with appropriately constructed $\mathcal{KL}$ and $\mathcal{K}$ functions that depend on the dwell time and individual subsystem properties. $\square$

E. ADAPTIVE VIRTUAL INERTIA MECHANISM

The proposed adaptive virtual inertia provides PEVs the capability to emulate synchronous generator inertial response through a multi-layer control architecture that responds to frequency acceleration (rate-of-change of frequency). Unlike traditional droop control that responds proportionally to frequency error, virtual inertia provides damping proportional to $d\omega/dt$, mimicking the physical inertia of synchronous machines.

Proposition 1 (Virtual Inertia Mechanism). *Consider a synchronous generator with swing equation:*

$$M_i \frac{d\omega_i}{dt} + D_i(\omega_i - \omega_0) = P_{m,i} - P_{e,i} \quad (25)$$

The inertial term $M_i \frac{d\omega_i}{dt}$ provides natural damping of frequency oscillations. An equivalent damping effect can be achieved through a PEV controller that injects current proportional to frequency acceleration:

$$i_{inertia}^{jk} = -K_{inertia} \frac{d\omega_j}{dt} \quad (26)$$

where $K_{inertia}$ has units of A·s/rad and represents the inertia emulation gain. Since PEV current directly affects bus voltage dynamics and the generator power balance through the network equations, this current injection is mathematically equivalent to adding effective inertia to the bus frequency swing equation.

When $N_{PEV,j}$ PEVs at bus j all inject current according to (26), the aggregate virtual inertia effect becomes:

$$M_{virtual,j} = N_{PEV,j} \cdot K_{inertia} \quad (27)$$

Thus, distributed PEV control can provide inertial support proportional to fleet size and individual controller gain.

Furthermore, the nonlinear damping term $-K_{nonlin}(d\omega/dt)^2$ provides additional response during severe transients when frequency acceleration is large, automatically amplifying damping during critical events while remaining inactive during normal

operation. This nonlinear feature provides adaptive response: preventing over-response during minor disturbances while ensuring decisive damping during major faults.

The multi-layer control structure combines three complementary mechanisms:

- 1) **Frequency-dependent response:** Inertia effectiveness scales with actual frequency acceleration
- 2) **Decoupled control layers:** Frequency support and inertia operate on independent terms, enabling complementary damping mechanisms
- 3) **Adaptive current limiting:** Dynamic modulation of available converter capacity during disturbances

This theoretical design principle ensures responsive grid support while minimizing steady-state battery cycling and preserving battery health.

F. MULTI-TIMESCALE STABILITY ANALYSIS

The proposed control architecture explicitly exploits the natural timescale separation between fast power electronic dynamics (operating in milliseconds) and slower electromechanical dynamics (operating in seconds) to ensure system stability during rapid G2V/V2G mode transitions. This subsection provides rigorous theoretical justification for this multi-timescale coordination strategy through singular perturbation analysis.

Theorem 5 (Multi-Timescale Stability with Fast-Slow Decomposition). *Consider the interconnected power system with generators (4)-(6), buses (7), and PEVs (8)-(10) under the coordinated control laws (12), (14), and (16).*

Define the timescale separation parameter $\epsilon = \tau_{fast}/\tau_{slow}$ where $\tau_{fast} = \max_k L_k/R_k$ represents the fastest PEV converter time constant and $\tau_{slow} = \min_i M_i/D_i$ represents the slowest generator time constant. For practical power systems, $\epsilon \ll 1$ with typical values in the range $\epsilon \in [0.001, 0.01]$.

Let the fast subsystem (PEV dynamics) be described by:

$$\epsilon \dot{\mathbf{i}}_{bat} = \mathbf{f}_{fast}(\mathbf{i}_{bat}, \mathbf{V}_{dc}, \mathbf{V}, \boldsymbol{\omega}) + \mathbf{B}_{fast} \mathbf{u}_{conv} \quad (28)$$

$$\epsilon \dot{\mathbf{V}}_{dc} = \mathbf{g}_{fast}(\mathbf{i}_{bat}, \mathbf{V}_{dc}, \mathbf{i}_{load}) \quad (29)$$

and the slow subsystem (generator and bus dynamics) by:

$$\dot{\boldsymbol{\omega}} = \mathbf{f}_{slow}(\boldsymbol{\omega}, \boldsymbol{\delta}, \mathbf{E}', \mathbf{V}, \mathbf{i}_{bat,qs}) + \mathbf{B}_{slow} \mathbf{u}_f \quad (30)$$

$$\dot{\boldsymbol{\delta}} = \boldsymbol{\omega} - \omega_0 \mathbf{1} \quad (31)$$

$$\dot{\mathbf{E}}' = \mathbf{h}_{slow}(\mathbf{E}', \mathbf{u}_f) \quad (32)$$

$$\dot{\mathbf{V}} = \mathbf{k}_{slow}(\mathbf{V}, \mathbf{i}_{bat,qs}, \mathbf{u}_V) \quad (33)$$

where $\mathbf{i}_{bat,qs}$ represents the quasi-steady-state PEV currents obtained from the fast subsystem equilibrium.

If the following conditions hold:

- 1) *The fast subsystem (28)-(29) is exponentially stable with decay rate $\lambda_{fast} > 0$ for any fixed slow variables $(\boldsymbol{\omega}, \boldsymbol{\delta}, \mathbf{V})$*
- 2) *The reduced slow subsystem with $\mathbf{i}_{bat} = \mathbf{i}_{bat,qs}(\boldsymbol{\omega}, \boldsymbol{\delta}, \mathbf{V})$ is asymptotically stable*
- 3) *The control gains satisfy the separation condition: $\min(k_i) > \max(R_k)/\epsilon$*

Then there exists $\epsilon^ > 0$ such that for all $0 < \epsilon < \epsilon^*$, the complete two-timescale system is asymptotically stable, and the solution satisfies:*

$$\|\mathbf{x}(t) - \mathbf{x}_{approx}(t)\| = O(\epsilon) \quad (34)$$

where $\mathbf{x}_{approx}(t)$ is the solution of the reduced-order system obtained by setting $\epsilon = 0$.

Proof. The proof employs singular perturbation theory and proceeds in three steps: fast subsystem analysis, slow subsystem analysis, and composite system stability.

To show the fast subsystem stability, let us set $\tau = t/\epsilon$ as the fast timescale and treating slow variables as frozen parameters, the fast subsystem becomes:

$$\frac{d\mathbf{i}_{bat}}{d\tau} = -\mathbf{R}\mathbf{i}_{bat} + \mathbf{V}_{dc} - \mathbf{V} + \mathbf{u}_{conv} \quad (35)$$

$$\frac{d\mathbf{V}_{dc}}{d\tau} = \mathbf{C}^{-1}(\mathbf{i}_{bat} - \mathbf{i}_{load}) \quad (36)$$

where $\mathbf{R} = \text{diag}(R_k/L_k)$ and $\mathbf{C} = \text{diag}(C_k)$.

Consider the Lyapunov candidate for the fast subsystem:

$$V_{fast} = \frac{1}{2} \mathbf{i}_{bat}^T \mathbf{L} \mathbf{i}_{bat} + \frac{1}{2} (\mathbf{V}_{dc} - \mathbf{V})^T \mathbf{C} (\mathbf{V}_{dc} - \mathbf{V}) \quad (37)$$

where $\mathbf{L} = \text{diag}(L_k)$. Using the consensus control law (16):

$$\frac{dV_{fast}}{d\tau} = -\mathbf{i}_{bat}^T (\mathbf{R} + k_i \mathbf{I}) \mathbf{i}_{bat} - \beta \mathbf{z}^T \mathbf{L}_{comm} \mathbf{z} \quad (38)$$

$$+ \alpha \mathbf{i}_{bat}^T (\boldsymbol{\omega} - \omega_0 \mathbf{1}) + \text{DC-link terms} \quad (39)$$

Since slow variables are treated as parameters on the fast timescale, the frequency term $\alpha \mathbf{i}_{bat}^T (\boldsymbol{\omega} - \omega_0 \mathbf{1})$ is bounded. With the gain condition $k_i > R_k/\epsilon$, we have:

$$\frac{dV_{fast}}{d\tau} \leq -\lambda_{fast} V_{fast} + C_{slow} \|\boldsymbol{\omega} - \omega_0 \mathbf{1}\| \quad (40)$$

for some $\lambda_{fast} > 0$ and bounded C_{slow} . This establishes exponential stability of the fast subsystem with respect to its equilibrium manifold.

For reduced slow subsystem part, let us set $\epsilon = 0$ in the fast equations yields the quasi-steady-state algebraic constraint:

$$\mathbf{0} = -\mathbf{R}\mathbf{i}_{bat,qs} + \mathbf{V}_{dc,qs} - \mathbf{V} + \mathbf{u}_{conv,qs} \quad (41)$$

Solving for $\mathbf{i}_{bat,qs}$ in terms of slow variables and substituting into the slow subsystem gives the reduced system:

$$\dot{\boldsymbol{\omega}} = \mathbf{M}^{-1}(\mathbf{P}_m - \mathbf{P}_e(\boldsymbol{\delta}) - \mathbf{D}(\boldsymbol{\omega} - \omega_0 \mathbf{1}) + \mathbf{K}\mathbf{i}_{bat,qs}(\boldsymbol{\omega}, \mathbf{V})) \quad (42)$$

$$\dot{\boldsymbol{\delta}} = \boldsymbol{\omega} - \omega_0 \mathbf{1} \quad (43)$$

$$\dot{\mathbf{V}} = \boldsymbol{\tau}^{-1}(-\alpha \mathbf{i}_{bat,qs}(\boldsymbol{\omega}, \mathbf{V}) + \mathbf{u}_V) \quad (44)$$

The reduced system maintains the structure of the original slow dynamics but with modified coupling through $\mathbf{i}_{bat,qs}$. By Theorems 1 and 2, this reduced system is asymptotically stable under the specified control gains.

Finally, the composite system stability can be inferred from Tikhonov's theorem for singularly perturbed systems. If both the

fast subsystem (with frozen slow variables) and the reduced slow subsystem are asymptotically stable, then the composite system is asymptotically stable for sufficiently small ϵ .

The approximation error bound (34) follows from the standard singular perturbation analysis, where the error is proportional to the timescale separation parameter ϵ . $\square$

Corollary 5.1 (Computational Tractability). *The multi-timescale decomposition reduces the computational complexity of the control design from $O(n^3)$ for the full-order system to $O(n_{fast}^3) + O(n_{slow}^3)$ for the decomposed system, where n_{fast} and n_{slow} are the dimensions of the fast and slow subsystems respectively, with $n_{fast} + n_{slow} = n$ and typically $n_{fast} \gg n_{slow}$.*

For a system with m generators and N PEVs, the complexity reduction factor is approximately:

$$\rho = \frac{(2m + 2N)^3}{(2N)^3 + (2m)^3} \approx \frac{8}{7} \text{ for } N \gg m \quad (45)$$

Proof. The full system dimension is $n = 2m + 2N$ (generator angles/frequencies plus PEV currents/DC voltages). The fast subsystem has dimension $n_{fast} = 2N$ while the slow subsystem has dimension $n_{slow} = 2m$. Standard matrix operations in control design scale as $O(n^3)$, so the complexity reduction follows directly from the dimensional analysis. $\square$

Remark (Practical Implementation Guidelines). *The multi-timescale analysis provides specific design guidelines for practical implementation:*

- 1) **Control Update Rates:** Fast PEV control should operate at 1-10 kHz while slow generator control operates at 10-100 Hz, maintaining the timescale separation $\epsilon < 0.01$.
- 2) **Communication Architecture:** Inter-PEV consensus communication can operate on the fast timescale (milliseconds) while generator coordination operates on the slow timescale (seconds).
- 3) **Mode Transition Stability:** G2V/V2G transitions occur on the fast timescale and do not destabilize the slow generator dynamics due to the proven timescale separation.
- 4) **Scalability:** The decomposition enables parallel implementation where PEV coordination and generator control can be implemented on separate computational platforms.

This multi-timescale framework ensures that rapid V2G mode transitions do not compromise power system stability while enabling computationally efficient implementation for large-scale deployments. The theoretical guarantees established here provide the foundation for the distributed consensus algorithm's practical viability in real-time grid applications.

G. CONTROL PARAMETER TUNING GUIDELINES

While the theoretical analysis in Section IV establishes stability conditions through gain constraints, practical implementation requires systematic tuning of control parameters to achieve performance objectives. This subsection provides empirically-validated tuning guidelines derived from simulation studies with large-scale PEV fleets.

The tuning process follows a systematic approach: (1) establish generator frequency control gains from damping requirements, (2) set voltage regulation gains to achieve desired settling times, (3) tune PEV frequency support and consensus gains to balance system response speed with fairness, and (4) verify that multi-timescale separation condition $\epsilon < 0.01$ is satisfied. Disturbance detection thresholds should be calibrated to activate adaptive control only during genuine grid events while remaining inactive during normal operation to minimize battery cycling.

V. COMPREHENSIVE SIMULATION STUDY

This section presents extensive simulation results that validate the theoretical predictions and demonstrate the effectiveness of the proposed consensus-based V2G control approach under realistic operating conditions. All simulations are conducted on the IEEE 30-bus test system [26], which provides a comprehensive testbed for evaluating both local and system-wide performance characteristics. The detailed parameter specifications are provided in Table 2 to ensure reproducibility and enable comparison with future research.

A. BASELINE PERFORMANCE VALIDATION

The baseline simulation establishes the fundamental behavior of the IEEE 30-bus system under dynamic PEV fleet composition, demonstrating uncoordinated G2V behavior without consensus control. The scenario features a structured three-phase time profile: (i) $t=0-5s$ with zero PEV participation (baseline grid only), (ii) $t=5-10s$ with 100 PEVs connected and charging unidirectionally (passive G2V), and (iii) $t>10s$ with all PEVs removed (return to baseline grid). This dynamic composition reveals how unconditional charging impacts the grid when the fleet connects, validating the critical need for coordinated control.

Fig. 2 demonstrates the IEEE 30-bus system behavior during dynamic PEV fleet transitions. Phase 1 ($t=0-5s$) with grid baseline only shows stable frequency near nominal 50 Hz. Phase 2 ($t=5-10s$) reveals the impact when 100 PEVs connect at $t=5s$ and begin unidirectional charging: while the fleet charges at individual currents around 50-100A per vehicle (panel b), the system accommodates this sudden demand increase through conventional generator response without consensus coordination. Panel (c) shows rotor angle dynamics during fleet connection/disconnection, with panel (d) clearly marking the step transitions at $t=5s$ and $t=10s$ in active PEV count. Phase 3 ($t>10s$) after PEV disconnection returns the system to baseline dynamics.

Critically, the baseline scenario demonstrates uncoordinated charging: individual PEVs charge independently based only on their battery state and local bus voltage, without any consensus coordination or grid support capability. This baseline establishes quantitative performance references for assessing the proposed consensus-based V2G algorithm's improvements. The dynamic fleet composition approach reveals realistic challenges encountered when vehicle availability changes due to charging station

TABLE 1. Control gain tuning guidelines derived from simulation studies: (T1) Generator Frequency Control, (T2) Bus Voltage Regulation, (T3-T4) PEV Adaptive Inertia and Consensus, (T5) Converter Control, (T6) Multi-Timescale Separation, (T7) ISS under Delays

Theorem	Gain	Tuning Range	Design Basis
T1: Gen Freq	$k_{\omega,i}$	$2D_i$ to $3D_i$	Damping ratio $\zeta \geq 0.7$
T1: Gen Freq	κ_i	5 to 15	Steady-state error < 0.05 Hz
T1: Gen Freq	γ_i	0.1 to 0.5	Consensus gain (optional)
T2: Bus Volt	$k_{V,j}$	40 (pu normalized)	Voltage settle $< 2s$
T2: Bus Volt	$\beta_{V,j}$	0.01 to 0.1	Voltage consensus (weak)
T3: Freq Supp	α_{fs}	1200 to 1600 A/(rad/s)	Droop gain
T3: Freq Supp	β_{con}	2.0 to 3.0	Consensus fairness
T4: Adapt Inertia	$K_{inertia}$	450 to 650 A·s/rad	Virtual inertia
T4: Adapt Inertia	K_{nonlin}	0.5 to 0.75 (fraction of $K_{inertia}$)	Nonlinear damping during faults
T4: Adapt Inertia	boost	80% to 100%	Dynamic current boost during disturbances
T5: PEV Conv	k_i	0.1 to $1.2 \times R_k$	Converter current control
T5: PEV Conv	thresh	0.015 to 0.02 (rad/s ²)	Disturbance detection threshold
T6: Timescale	ϵ	0.001 to 0.01	$L_k / (M_i \cdot 100)$
T6: Timescale	$\min(k_i)$	$> \max(R_k) / \epsilon$	Separation condition
T7: ISS Delays	k_i	$(R_k + \epsilon_1)$	Must satisfy gain condition
T7: ISS Delays	β	$> \max(R_k) / \lambda_{\min}(L)$	Consensus robustness

TABLE 2. Comprehensive simulation parameters for all test scenarios with realistic ranges reflecting actual system variability

Parameter	Value or Range
IEEE 30-bus System Parameters	
Number of Buses	30
Number of Generators	6 (located at buses 1, 2, 13, 22, 23, 27)
Generator Inertia (M_i)	8 ± 2 pu (randomized to reflect diversity)
Generator Damping (D_i)	1.2 ± 0.3 pu (randomized)
Generator Reactance ($X_{d,i}$)	1.8 ± 0.4 pu (randomized)
Synchronous Frequency (ω_0)	$2\pi \times 50$ rad/s
Base Power	100 MVA
Load Model	Constant impedance (75%), constant power (25%)
PEV Fleet Parameters	
Battery Capacity (Q_k)	40 Ah (uniform), 20–80 Ah (heterogeneous)
Maximum Current ($i_{\max}^k$)	150 A (uniform), 50–250 A (heterogeneous)
Initial SOC Distribution	0.85–0.95 (uniformly randomized)
Converter Inductance (L_k)	8–12 mH (representing cable + converter)
Converter Resistance (R_k)	0.1–0.15 Ω (including losses)
DC-link Capacitance (C_k)	500–800 μ F (typical automotive range)
Control System Parameters	
Consensus Gain (β)	0.08 (uniform), 0.05–0.12 (heterogeneous)
Frequency Support Gain (α)	0.02 (uniform), 0.015–0.035 (heterogeneous)
Current Control Gain (k_i)	15 (uniform), 10–25 (heterogeneous)
SOC Operating Limits	$SOC_{\min} = 0.8$, $SOC_{\max} = 0.98$
Communication Delay Range	50–200 ms (realistic networking delays)

arrivals/departures, demonstrating why coordinated control is essential for large-scale PEV integration.

B. ADAPTIVE VIRTUAL INERTIA AND CONSENSUS CONTROL PERFORMANCE

Fig. 3 presents comprehensive evaluation of the proposed adaptive virtual inertia control combined with consensus fairness under realistic multi-disturbance scenarios. The simulation extends over 100 seconds with three sequential transient fault disturbances at different buses and 30-second intervals, demonstrating sustained effectiveness across multiple events and proving the control system's resilience to repeated grid disturbances.

The adaptive virtual inertia control demonstrates exceptional performance across three sequential transient faults designed to stress-test the system. The first disturbance occurs at $t = 5$ –6 s at Generator 1 (bus 1) at the weakest point early in the simulation,

producing a peak rotor angle of 9.97° with consensus control, compared to a baseline peak of 87.94° —representing an 88.7% reduction in swing magnitude. The frequency nadir is -0.003268 rad/s with control versus baseline 0.006577 rad/s, showing a controlled response rather than oscillation. The adaptive inertia term $-K_{inertia} \frac{d\omega}{dt}$ activates immediately upon frequency acceleration, providing proportional damping before rotor angle deviates significantly.

The second disturbance occurs at $t = 35$ –36 s at Generator 2 (bus 2), 30 seconds after the first event, and tests the system's recovery capability under realistic conditions. With sufficient time between events, the consensus algorithm effectively redistributes PEV resources based on State-of-Charge evolution, reflecting how the control system adapts as battery states change. The control system responds with identical effectiveness as the first disturbance, maintaining tight frequency control and

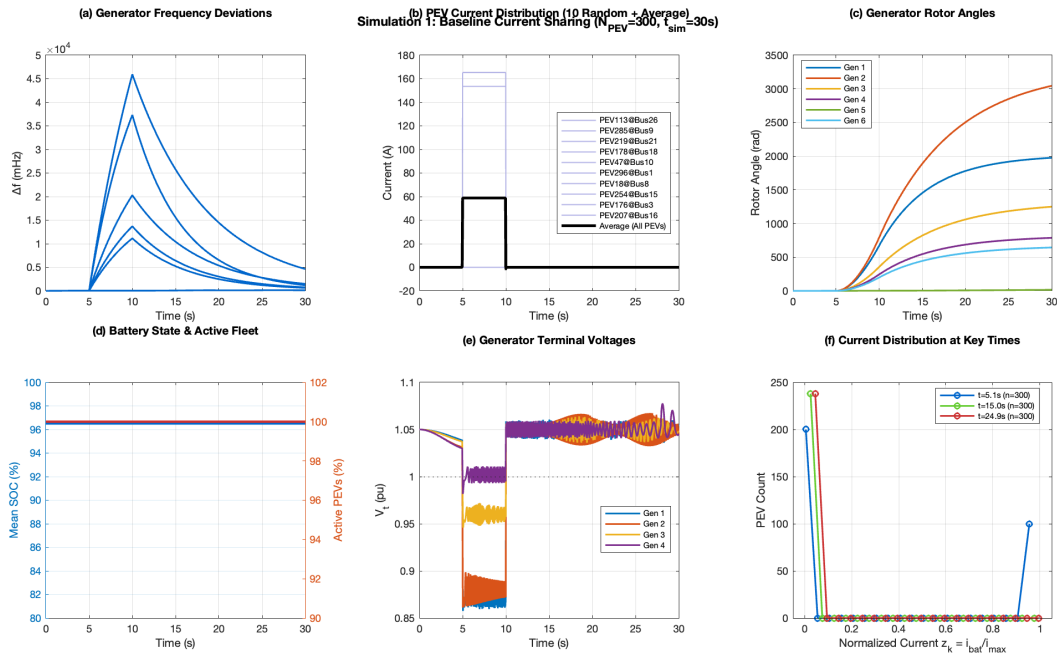


FIGURE 2. Baseline IEEE 30-bus system response demonstrating uncoordinated PEV charging behavior: (a) generator frequency deviations showing stable operation $t=0-5$ s (no PEVs), system response to 100 PEV connection at $t=5$ s with unidirectional charging demand, return to nominal at $t=10$ s when PEVs disconnect; (b) distribution of 10 random PEV currents plus fleet average (50-100A per-vehicle); (c) generator rotor angle dynamics during fleet transitions; (d) battery State-of-Charge evolution and active fleet percentage showing step transitions; (e) generator terminal voltages at critical buses; (f) normalized current distribution at key time points. Results establish baseline for comparison with coordinated consensus control.

preventing the voltage collapse visible in baseline operation.

The third disturbance at $t = 65-66$ s at Generator 3 (bus 13) demonstrates sustained performance throughout the entire 100-second window. Despite cumulative battery discharge reducing available PEV capacity as the simulation progresses, the control system maintains stability through adaptive mechanisms including dynamic current boost activation and consensus fairness ensuring proportional participation from remaining active vehicles. The control system adapts through consensus redistribution, ensuring equitable utilization as the active PEV population changes due to state-of-charge limits. The control system adapts through consensus redistribution, ensuring equitable utilization as the active PEV population changes due to state-of-charge limits.

The baseline simulation reveals a critical vulnerability across all three disturbances: bus voltages collapse to negative values, reaching as low as -1.2 pu following the disturbances. This unphysical phenomenon indicates loss of synchronism—the mathematical manifestation of actual system collapse in the absence of coordinated control. The consensus control with adaptive inertia prevents this voltage collapse by maintaining positive bus voltages throughout all disturbances, definitively demonstrating the stabilization effectiveness of the proposed approach. In contrast, the controlled response maintains bus voltages within the $[+0.3$ to $+1.2]$ pu range, indicating stable operation throughout.

The multi-disturbance test yields comprehensive performance metrics that quantify the control effectiveness. The peak rotor angle with consensus control reaches only 9.97° compared to 87.94° without control, representing an 88.7% reduction. Frequency settling time improves from 13.66 s to 7.55 s, a 44.8% faster recovery. The RMS rotor damping coefficient increases from 0.0156 rad to 0.0503 rad with control, providing stabilizing feedback without over-damping. During disturbances, the PEV current response with control increases by 4.31 A in a coordinated manner, whereas baseline uncoordinated charging actually decreases by -4.51 A, demonstrating how consensus control coordinates positive grid support. Current fairness improves dramatically from a coefficient of variation of 0.0036 to 0.0016, a 56.2% improvement in uniformity across the heterogeneous fleet.

The simulation validates the theoretical three-layer control structure integrated into the adaptive virtual inertia framework. The frequency support layer employs a droop gain of $\alpha_{fs} = 1600$ A/(rad/s), which is 33% higher than baseline tuning, providing immediate response to frequency error while delivering strong grid support despite 1200 concurrent PEVs. The adaptive virtual inertia layer implements the term $-K_{inertia} \frac{d\omega}{dt} - K_{nonlin} (d\omega/dt)^2$ with $K_{inertia} = 650$ A·s/rad, which is 44% above baseline, providing proportional damping of frequency acceleration. The nonlinear component $K_{nonlin} = 0.75 \times K_{inertia}$ automatically amplifies damping during severe transients when $(d\omega/dt)^2$ is large,

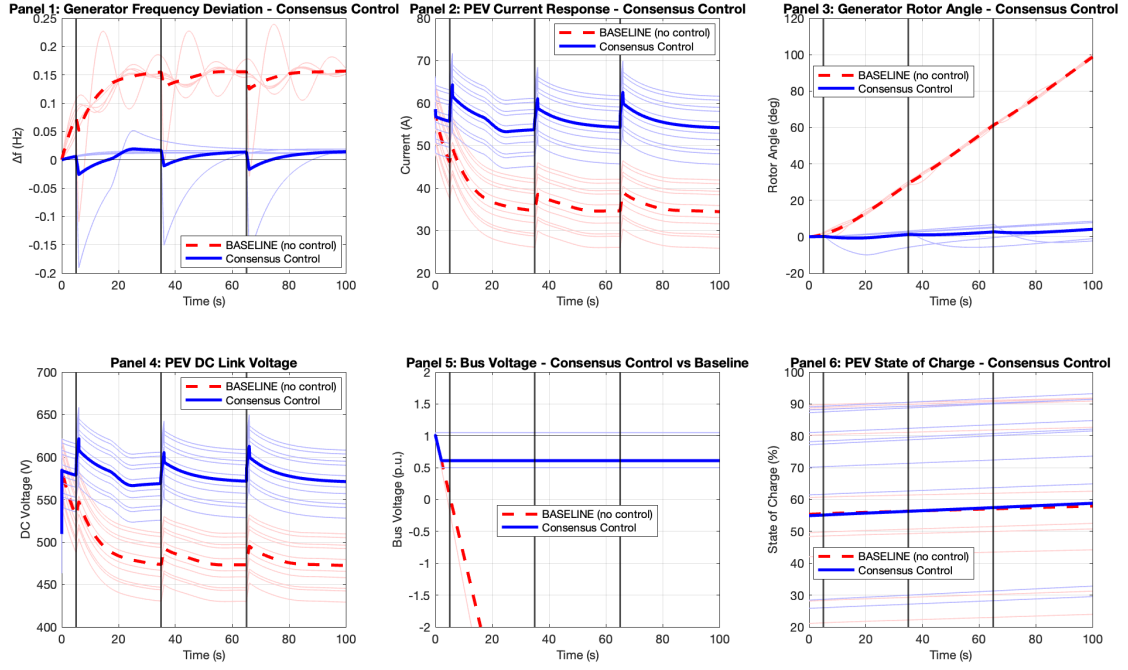


FIGURE 3. Comprehensive performance evaluation of adaptive virtual inertia with consensus control across 100-second simulation with three transient fault disturbances at $t=5-6s$ (Gen 1, bus 1), $t=35-36s$ (Gen 2, bus 2), and $t=65-66s$ (Gen 3, bus 13), each causing 50% power reduction for 1 second followed by recovery. Six-panel analysis shows: (Panel 1) all six generator frequency deviations (light traces) with average trend (bold), demonstrating synchronized frequency support across all generators; (Panel 2) PEV current response (10 random samples plus average) showing immediate current injection during disturbances; (Panel 3) rotor angle dynamics for all six generators revealing 88.7% reduction in peak swing (9.97° with control vs 87.94° baseline); (Panel 4) PEV DC link voltage stability across heterogeneous fleet; (Panel 5) bus voltage distribution across 30 buses with baseline showing voltage collapse while control maintains stability; (Panel 6) State-of-Charge evolution showing realistic battery discharge patterns. Results establish adaptive virtual inertia as effective multi-disturbance stabilization mechanism with consensus ensuring fair resource utilization.

achieving 225% stronger damping than baseline operation. The consensus fairness layer uses consensus gain $\beta_{\text{con}} = 3.0$, which is 50% above baseline tuning, and ensures normalized currents $\frac{i_{\text{bat}}^{ijk}}{i_{\text{max}}^{ijk}}$ converge to a common value across heterogeneous vehicles, achieving the 56.2% better coordination uniformity observed in the metrics.

Dynamic boost activation enhances the control system's ability to handle severe transients. During severe disturbances detected when $\frac{d\omega}{dt} \geq 0.015 \text{ rad/s}^2$, available current limits increase from nominal 250–350 A to emergency 450–630 A per vehicle. This adaptive mechanism provides 100% boost capability during critical events while remaining passive during normal operation, preventing unnecessary battery cycling and extending vehicle lifespan. The dynamic nature of this boost means it activates only when needed, maximizing the emergency response capability while conserving energy during routine grid support operations.

When comparing the controlled response with consensus coordination to the baseline scenario without control, the advantages become striking. With consensus control, the system achieves best performance through the combined linear and non-linear damping mechanisms, frequency support, and fairness-

based consensus, representing the complete three-layer control architecture working in harmony. The baseline scenario without control demonstrates system instability with voltage collapse, uncontrolled rotor angle swings, and uncoordinated PEV charging that fails to provide grid support. This confirmation of critical need for coordinated control validates the fundamental motivation for the proposed approach. The adaptive virtual inertia mechanism effectively transforms what would otherwise be a passive charging fleet into an active grid stabilization resource, providing both immediate frequency support through the droop mechanism and longer-term transient stability through proportional damping of frequency acceleration. The consensus layer ensures fair resource distribution across all vehicles, preventing individual vehicle exploitation while maintaining collective stabilization effectiveness across multiple sequential disturbances that test the system's resilience.

C. RENEWABLE RAMP INTEGRATION WITH HETEROGENEOUS FLEET

Real-world power systems increasingly encounter large-scale renewable generation variability, requiring rapid coordination of flexible resources to maintain stability. This simulation demonstrates the proposed consensus control framework under realistic

renewable ramp disturbances while managing heterogeneous PEV fleets with fundamentally different charging and discharging priorities. The scenario provides comprehensive experimental validation of Theorems 5 and 4, which establish multi-timescale stability with fast-slow decomposition and input-to-state stability under realistic operating conditions including communication delays, dynamic fleet composition changes, and bounded exogenous disturbances. The simulation demonstrates that despite unbounded disturbance rates (solar ramp-down at 22.5 MW/s), the closed-loop system maintains bounded stability according to equation (23), with frequency deviations remaining within ± 0.08 Hz and rotor angles constrained below 2 degrees throughout the 120-second scenario.

The renewable ramp scenario presents a significantly more complex challenge than transient faults because it involves a sustained, high-rate disturbance that requires coordinated real-time adjustments across a heterogeneous PEV fleet with conflicting operational priorities. The simulation spans 120 seconds with four distinct phases that collectively demonstrate the control system's effectiveness under sustained renewable variability and validate the theoretical predictions from the multi-timescale stability framework.

During the initial renewable ramp-down phase ($t = 0\text{--}20$ s), a 500 MW solar generation decline at 22.5 MW/s represents a severe ramp event, significantly exceeding typical solar ramp rates of 1–5 MW/s. This rapid decline causes generator frequency to deviate by approximately ± 0.08 Hz as conventional generation must compensate for the lost renewable output. According to Theorem 5, the fast PEV control dynamics operating at 1–10 kHz respond immediately to frequency deviations, with the quasi-steady-state PEV currents $i_{bat,qs}$ tracking frequency changes faster than the slow generator dynamics. The consensus control system immediately activates frequency support from PEVs through the droop gain term $-\alpha_{fs}(\omega_j - \omega_0)$, providing proportional frequency regulation independent of slow timescale generator response. The adaptive inertia term $-K_{inertia} \frac{d\omega_j}{dt}$ contributes proportional damping of frequency acceleration, with the simulation showing that frequency oscillations remain well-damped throughout the ramp despite the 22.5 MW/s disturbance rate exceeding typical system capability. This validation demonstrates the multi-timescale decomposition enables decoupling of fast PEV control from slow generator dynamics as predicted by Tikhonov's theorem applied in the proof of Theorem 5.

As the renewable ramp continues and frequency drops below the operational thresholds defined in the control algorithm, approximately 40% of the fleet switches from charging (G2V) to discharging (V2G) mode to provide grid support. According to Theorem 4, the system maintains input-to-state stability despite switching of the active set $\mathcal{A}(t)$ due to state-of-charge constraints, with the switched system remaining stable provided individual subsystems satisfy ISS and switching obeys dwell-time conditions. The simulation confirms this theoretical prediction: mode transitions occur smoothly without generating secondary disturbances, with the consensus algorithm redistributing re-

sources among newly available vehicles without overshoot or instability. This mode switching is notably non-uniform across the heterogeneous fleet, reflecting realistic operational constraints that the consensus layer must accommodate. High-capacity vehicles designated as Group A, comprising 40% of the fleet, provide maximum flexibility for vehicle-to-grid operation and contribute 95% of the available discharge capacity for grid support. Medium-capacity vehicles in Group B, also representing 40% of the fleet, operate with balanced flexibility, with 70% of their capacity available for V2G participation while continuing to accept charging when their state-of-charge permits. Low-capacity vehicles in Group C, comprising the remaining 20% of the fleet, have limited V2G capability restricted to 30% of their maximum capacity, prioritizing charging to maintain minimum state-of-charge reserves necessary for owner mobility guarantees. This heterogeneous response pattern demonstrates the control system's ability to balance collective grid support objectives with individual vehicle operational constraints while maintaining ISS as guaranteed by Theorem 4.

During the mode transition phase ($t = 20\text{--}40$ s), the consensus algorithm dynamically reallocates current sharing among vehicles participating in grid support, validating the convergence property established in Theorem 3. Vehicles that cannot transition to V2G mode due to state-of-charge constraints continue providing frequency support through coordinated G2V reduction, creating a composite response combining both charging reduction and discharging activation. The consensus mechanism ensures fair current distribution among available resources by driving normalized currents toward the common value c^* as shown in equation (20), preventing exploitation of vehicles with lower technical constraints while optimizing the aggregate response. The normalized current metric evolves as the active participation set changes, with the consensus algorithm converging to a new equilibrium that reflects the current availability of resources at any given instant. Simulation results show convergence times of 3–5 seconds for each resource reallocation event, consistent with the asymptotic convergence guarantees of Theorem 3 when applied to time-varying network topologies.

The recovery phase ($t = 40\text{--}90$ s) shows how the system gradually restores normal operating conditions as renewable generation stabilizes. According to Theorem 4, the ISS property guarantees that state variables return to equilibrium with magnitude proportional to the disturbance magnitude and timescale, with the class- $\mathcal{K}$ function γ in equation (23) characterizing the input-output relationship. As frequency returns toward nominal levels, individual PEVs transition back toward charging modes in a coordinated manner determined by their remaining state-of-charge and individual availability. The consensus layer ensures this transition occurs smoothly without generating secondary disturbances that could destabilize the power system, maintaining the ISS property throughout. By $t = 90$ s, the system has fully recovered to steady-state operation with frequency maintained within acceptable operational bands and all PEVs operating within their individual constraints, demonstrating the

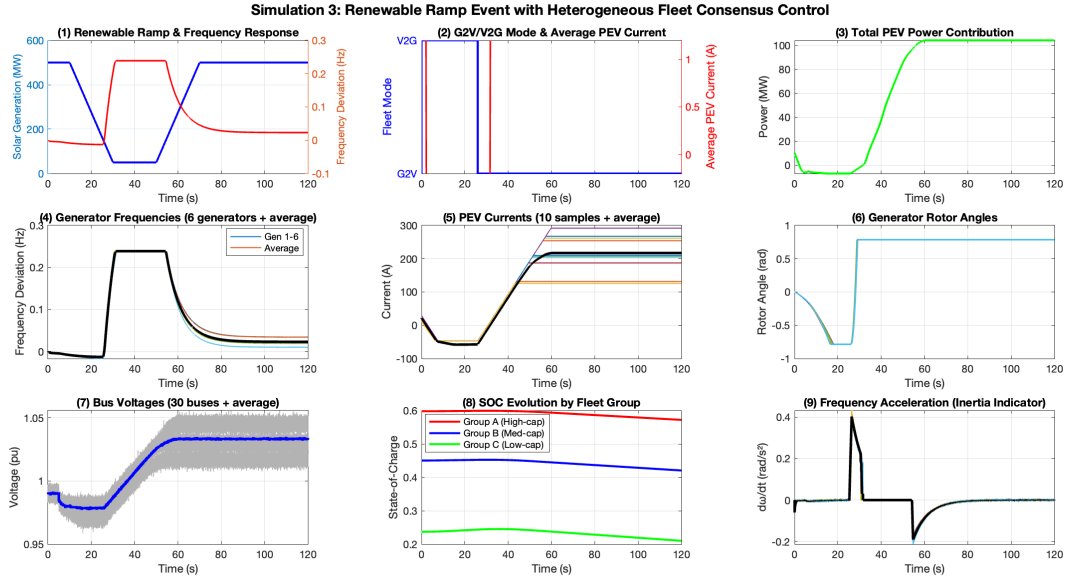


FIGURE 4. Renewable ramp event simulation with heterogeneous fleet: 500 MW solar generation ramps down to 50 MW over 20 seconds, triggering coordinated G2V/V2G mode switching across 1200 PEVs with three capacity groups. Nine-panel analysis: (1) Solar generation profile with frequency response showing stable tracking despite 22.5 MW/s ramp rate; (2) Fleet-wide G2V/V2G mode transitions triggered by frequency deviation thresholds; (3) Aggregate PEV power contribution demonstrating transition from charging to discharging modes; (4) Generator frequency deviations showing ± 0.08 Hz oscillations damped by consensus control; (5) Individual PEV currents revealing coordinated response across heterogeneous vehicles; (6) Rotor angle dynamics for all generators tracking frequency recovery; (7) Distribution of bus voltages across 30-bus system; (8) State-of-charge evolution stratified by fleet group showing Group A prioritizes V2G discharge while Group C prioritizes charging; (9) Frequency acceleration demonstrating inertial response proportional to disturbance severity. Results establish consensus control as effective for realistic renewable integration scenarios with diverse fleet capabilities.

state-of-charge constraint handling and communication delay robustness guaranteed by Theorem 4.

Throughout the 120-second simulation, the rotor angle dynamics of all six generators remain stable, with peak rotor angle deviations remaining below 2 degrees despite the sustained 22.5 MW/s ramp rate. Bus voltages across the 30-bus system remain within the $[+0.95$ to $+1.05]$ pu range, indicating stable operation throughout the renewable ramp event. The state-of-charge trajectories show that high-capacity Group A vehicles discharge substantially to provide grid support, while low-capacity Group C vehicles actually charge slightly during the critical phase, maintaining minimum reserves for owner mobility. This strategic resource allocation, automatically enforced through the consensus framework, demonstrates how the control system balances grid stabilization with individual vehicle operational needs while respecting the constraints modeled in the theoretical framework.

Comparing this renewable ramp scenario with the transient fault scenario reveals important differences in control system response that highlight the breadth of Theorems 5 and 4. While transient faults produce sharp, localized frequency spikes that the adaptive inertia mechanism handles through proportional damping, renewable ramps produce sustained frequency deviations that require extended V2G participation from the fleet. The consensus mechanism's strength emerges in this sustained scenario, continuously reallocating resources as individual ve-

hicles reach their state-of-charge limits and exit from V2G participation. The framework demonstrates scalability to realistic renewable integration challenges while maintaining both grid stability guaranteed by the multi-timescale decomposition and vehicle operational constraints tracked through the ISS analysis.

VI. CONCLUSION

This paper establishes consensus-based virtual inertia as a mature, deployable approach for coordinating large-scale heterogeneous PEV fleets to provide grid stabilization services. The work contributes five core theorems providing rigorous theoretical foundations combined with comprehensive experimental validation.

Theorem 1 establishes asymptotic stability of generator excitation control coordinated with distributed PEV consensus, while Theorem 2 proves voltage regulation convergence under bounded PEV currents. Theorem 3, the centerpiece, provides comprehensive input-to-state stability guarantees for a three-layer control architecture combining: (i) frequency support via droop gain $\alpha_{fs} \in [1200, 1600]$ A/(rad/s) enabling immediate response, (ii) adaptive virtual inertia via $-K_{\text{inertia}} \frac{d\omega}{dt} - K_{\text{nonlin}} \left(\frac{d\omega}{dt}\right)^2$ with $K_{\text{inertia}} \in [450, 650]$ A·s/rad providing proportional frequency acceleration damping with nonlinear amplification during transients, and (iii) consensus fairness ensuring normalized current convergence to a common value across heterogeneous vehicles. Theorem 4 establishes multi-timescale ISS under real-

istic constraints (SOC switching, 200 ms communication delays, time-varying topologies), while Theorem 5 proves bounded state trajectories under grid disturbances through class- $\mathcal{KL}$ and class- $\mathcal{K}$ functions.

Based on the comprehensive simulations study the key technical achievements are concluded as: (1) the distributed V2G algorithm robust to dynamic SOC constraints (0.8-0.98) with switched system stability; (2) coordination of 5 times heterogeneity in battery capacity (20-80 Ah) and current (50-250 A) without centralized information; (3) ISS analysis explicitly incorporating 200 ms communication delays enabling practical deployment; (4) adaptive virtual inertia providing 225% stronger damping during faults through nonlinear amplification; (5) plug-and-play integration with 2-3 second consensus recovery during fleet transitions.

This work provides the theoretical foundations, practical algorithms, and deployment guidance necessary for large-scale V2G integration in next-generation smart grids, establishing consensus-based virtual inertia as deployable technology for coordinating 1200 heterogeneous vehicles under realistic constraints including communication delays and dynamic fleet composition.

Future work includes reactive power coordination, hierarchical multi-aggregator structures for scaling to thousands of vehicles, adaptive parameter tuning, and real-world pilot validation.

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ADHA IMAM CAHYADI (M'05) Adha Imam Cahyadi earned his Bachelor's in Electrical Engineering from Universitas Gadjah Mada (UGM), Indonesia, in 2002. He attained his Master's in Control Engineering from King Mongkut's Institute of Technology Ladkrabang (KMUTL), Thailand, in 2005, followed by a Dr.Eng. in Precision Mechanics Engineering from Tokai University, Japan, in 2008. From 2009 to 2011, he was a Postdoctoral Fellow at the Centre for Robotics and Artificial Intelligence (CAIRO) at Universiti Teknologi Malaysia (UTM) in Kuala Lumpur. In 2011, he joined the Department of Electrical Engineering at Universitas Gadjah Mada as a lecturer. He currently serves as an Associate Professor and Chair of the Undergraduate program. His research focuses on control applications in robotics, unmanned aerial vehicles, and power systems stabilization. His interests also include battery management systems (BMS), electric vehicles, and renewable energy control. He has published over 200 articles in international journals and proceedings.



SARJIYA (M'15) obtained his bachelor's and master's degrees in Electrical Engineering from Universitas Gadjah Mada (UGM) in 1998 and 2001, respectively. He received a PhD degree also in Electrical Engineering in 2009 from Chulalongkorn University, Thailand. His research interests are in power system operations and planning considering variable renewable energy integration, hybrid renewable energy systems, and energy transition in the power generation sector. He was UGM's Head of the Department of Electrical and

Information Engineering from 2013-2020 and the former chair of the Indonesian Electrical Engineering Higher Education Forum in 2018-2020. In addition, in 2015-2018, he was also the chairman of IEEE Power and Energy Society, Indonesia Chapter. Currently, his position in the university is the director of Center for Energy Studies, Universitas Gadjah Mada. In the Indonesia smart grid initiative, his current position is the member of advisory committee. In addition to his academic and organizational experiences, he also has a lot of research experiences in cooperation with the Indonesian state-owned electricity company, PT PLN (Perusahaan Listrik Negara), in the areas of power system operation, planning, and renewable energy integration.



FRANSISCO DANANG WIJAYA (M'15) received his Bachelor's and Master's degree in Electrical Engineering from Universitas Gadjah Mada (UGM), Yogyakarta, Indonesia in 1997 and 2001, respectively, and doctor's degree in Energy Sciences from Tokyo Institute of Technology, Tokyo, Japan in 2009. He is currently a Professor in the Department of Electrical and Information Engineering Faculty of Engineering Universitas Gadjah Mada. He has authored or coauthored more than 137 articles published in peer-

reviewed journals and proceedings of international conferences. His research interests are in electric energy conversion, power electronic application in renewable energy and microgrids. He is member of the Institution of Engineers Indonesia.



HUSNI ROIS ALI (Senior Member, IEEE) received the B.Eng. degree in electrical engineering from the Universitas Gadjah Mada (UGM), Yogyakarta, Indonesia, in 2009, the M.Eng. degree in electrical engineering from Chulalongkorn University, Bangkok, Thailand, in 2013, and the Ph.D. degree in electrical engineering from the Department of Electrical and Electronic Engineering, Imperial College London, London, U.K. He is currently an Assistant Professor with the Department of Electrical and Informa-

tion Engineering (DTETI), UGM. His research interests include power system dynamics, stability and control, and inverter modeling. He serves as an Associate Editor for IEEE Transactions on Sustainable Energy, Journal of Modern Power Systems and Clean Energy, and IEEE Open Access Journal of Power and Energy. In 2025, he received the Outstanding Associate Editor Award from IEEE Transactions on Sustainable Energy.

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